

Smart Temperature monitoring using BLYNK

Project Aim

To develop an IoT-based smart temperature monitoring and automatic fan control system using ESP32, DHT22, and Blynk Cloud.

Project Overview

This project uses an ESP32 and a DHT22 sensor to monitor temperature and humidity in real time. The ESP32 sends sensor data to the Blynk Cloud through Wi-Fi, allowing remote monitoring from a mobile or web dashboard. An LED connected to GPIO 13 acts as a fan indicator and automatically turns ON when the temperature exceeds 30°C and turns OFF when the temperature is 30°C or below.

Components Used

- ESP32 DevKit V1
- DHT22 Temperature and Humidity Sensor
- LED
- 220Ω Resistor
- Jumper Wires
- Blynk Cloud Account

Construction and Wiring Details

Component Component Pin ESP32 Pin

DHT22	VCC	3.3V
DHT22	SDA (DATA)	GPIO 4
DHT22	GND	GND
LED	Anode (+)	GPIO 13

Component Component Pin ESP32 Pin

LED Cathode (-) 220Ω Resistor → GND

Blynk Connection

Virtual Pin	Function
V0	Temperature Display
V1	Humidity Display
V2	Fan Status Display

Code Structure / Program Flow

START

|



Initialize ESP32

|



Connect to Wi-Fi

|

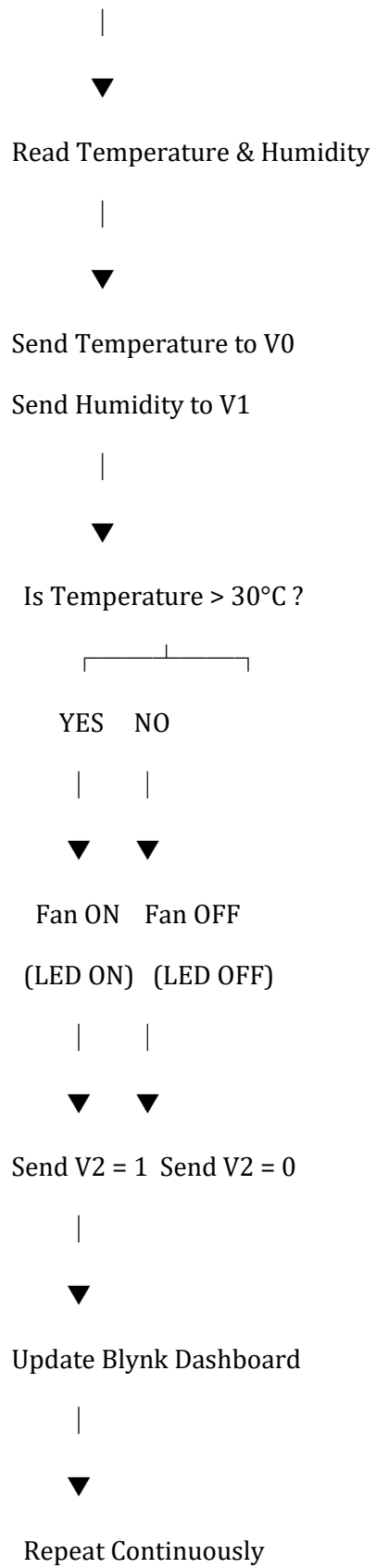


Connect to Blynk Cloud

|



Initialize DHT22



|



END END

Source Code

Diagram.jetson

```
{  
  "version": 1,  
  "author": "Aman Abdulla",  
  "editor": "wokwi",  
  "parts": [  
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": 0, "left": 0, "attrs": {} },  
    {  
      "type": "wokwi-led",  
      "id": "led1",  
      "top": -118.8,  
      "left": 186.2,  
      "attrs": { "color": "red" }  
    },  
    {  
      "type": "wokwi-resistor",  
      "id": "r1",  
      "top": -33.6,  
      "left": 181.85,  
      "rotate": 90,  
      "attrs": { "value": "1000" }  
    },  
  ],  
}
```

```

{
  "type": "wokwi-dht22",
  "id": "dht1",
  "top": -134.1,
  "left": -130.2,
  "attrs": { "temperature": "64.7" }
}
],
"connections": [
  [ "esp:TX", "$serialMonitor:RX", "", [] ],
  [ "esp:RX", "$serialMonitor:TX", "", [] ],
  [ "esp:GND.2", "led1:C", "black", [ "v-19.2", "h91.24" ] ],
  [ "led1:A", "r1:1", "green", [ "v0" ] ],
  [ "esp:GND.1", "dht1:GND", "black", [ "h-71.81" ] ],
  [ "dht1:VCC", "esp:3V3", "red", [ "v0" ] ],
  [ "esp:4", "dht1:SDA", "green", [ "h33.64", "v-182.4", "h-182.4", "v115.2", "h-48" ] ],
  [ "esp:13", "r1:2", "green", [ "h-23.81", "v67.2", "h240", "v-9.6" ] ]
],
"dependencies": {}
}

```

Esp32 Code

```

#define BLYNK_TEMPLATE_ID "TMPL3Vi3-17sH"

#define BLYNK_TEMPLATE_NAME "Smart Temperature Monitor"

#define BLYNK_AUTH_TOKEN "YFqi8ra_w_YIQLaKMGRiFM7FKwgxWVPF"

#include <WiFi.h>

```

```
#include <BlynkSimpleEsp32.h>

#include <DHT.h>

char ssid[] = "Wokwi-GUEST";
char pass[] = "";

#define DHTPIN 4

#define DHTTYPE DHT22

#define FAN_LED 13

DHT dht(DHTPIN, DHTTYPE);

BlynkTimer timer;

void sendSensor()
{
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    if (isnan(temperature) || isnan(humidity))
    {
        Serial.println("Failed to read DHT22!");
        return;
    }
}
```

```
Serial.print("Temperature: ");
```

```
Serial.print(temperature);
```

```
Serial.print(" °C ");
```

```
Serial.print("Humidity: ");
```

```
Serial.print(humidity);
```

```
Serial.println(" %");
```

```
Blynk.virtualWrite(V0, temperature);
```

```
Blynk.virtualWrite(V1, humidity);
```

```
if (temperature > 30)
```

```
{
```

```
    digitalWrite(FAN_LED, HIGH);
```

```
    Blynk.virtualWrite(V2, 1);
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(FAN_LED, LOW);
```

```
    Blynk.virtualWrite(V2, 0);
```

```
}
```

```
}
```

```
void setup()
```

```
{
```

```
    Serial.begin(115200);
```

```
pinMode(FAN_LED, OUTPUT);
```

```
dht.begin();
```

```
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
```

```
timer.setInterval(2000L, sendSensor);
```

```
}
```

```
void loop()
```

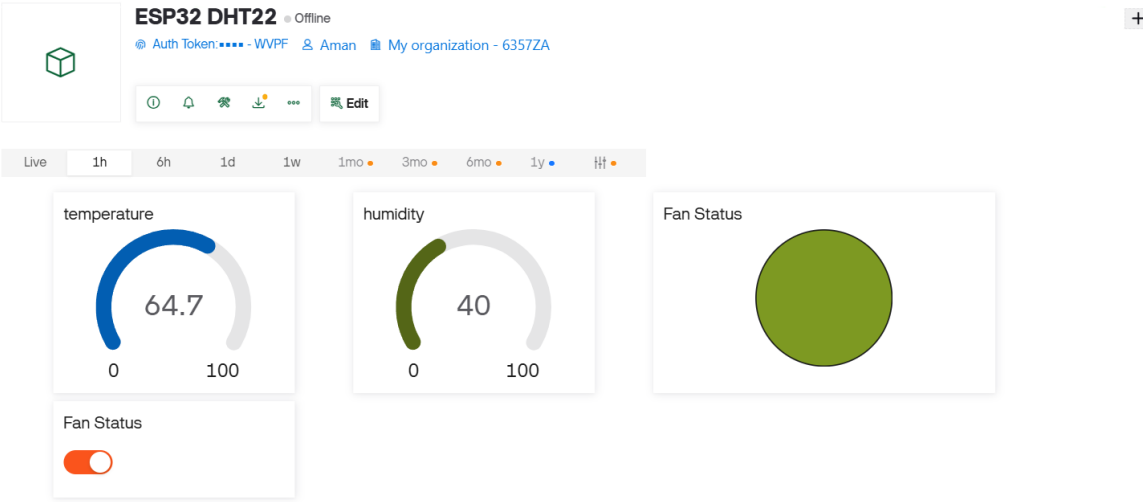
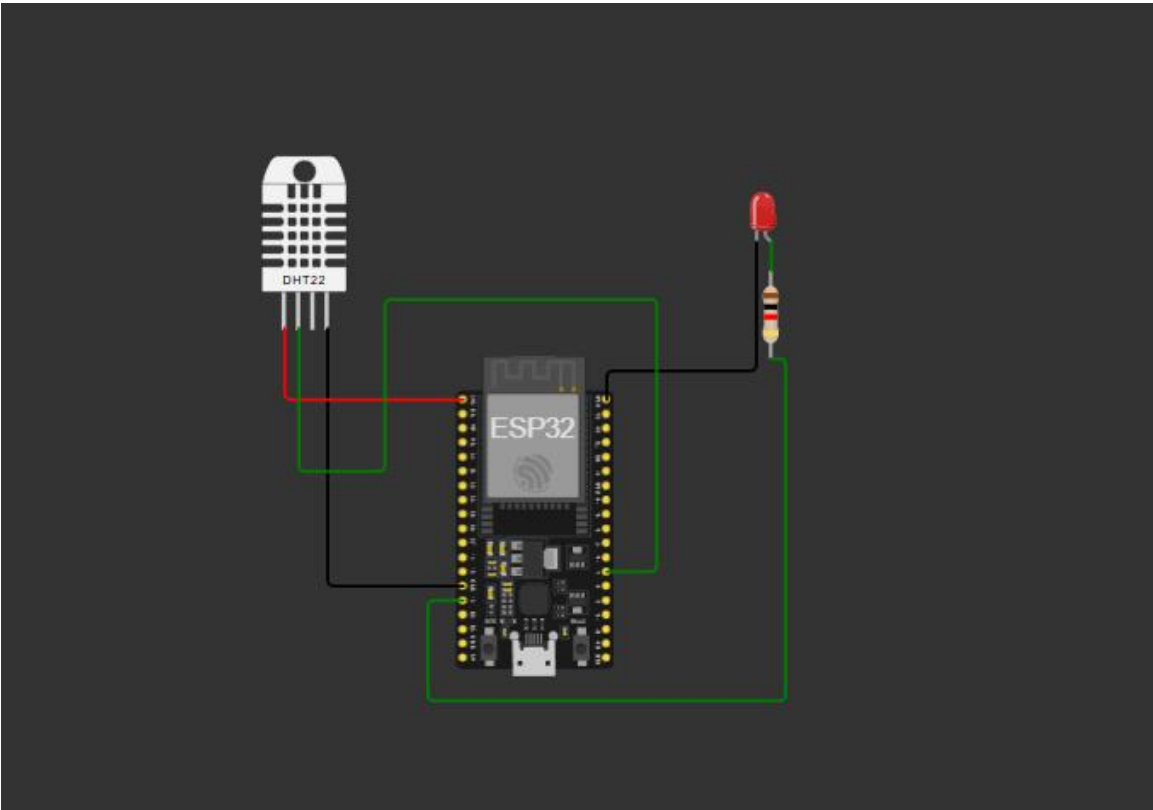
```
{
```

```
  Blynk.run();
```

```
  timer.run();
```

```
}
```


Project Photograph



Links

Google drive :

<https://drive.google.com/drive/folders/1tAx9U7EucOQ2EOWxABzoAs4SYYvp8Ubp?usp=sharing>

Wokwi : <https://wokwi.com/projects/466474552849638401>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>